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Department of Computer Science and Engineering (AI&ML)

Major ProjectReview-2(Semester VII)

Hierarchical Position Encoding for Structure-Sensitive Semantic Dense Retrieval

Group No : 18

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Outline of Presentation

- Problem Definition
- Introduction
- Goals and Objectives
- Motivation
- Literature Review (Atleast 10 to15 Papers in Tabular Format)
- Existing System & its Limitation
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- Methodology
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- Project Timeline (Work Completed,currect Stage ,Upcoming Activities)
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Problem Definition

- Dense retrieval systems embed documents into high-dimensional vector spaces where semantic proximity drives retrieval. The position of a token in that space is determined entirely by its sequence position — a single integer encoding linear distance from the start of the text.
- Every production embedding model uses this one-dimensional position signal. The result: the vector space has no axis for structural meaning — no dimension for depth in a hierarchy, no dimension for which section a token belongs to, no dimension for its relational role in the document.
- When a document has structure { a financial report, a legal contract, a technical specification } the same sentence can mean opposite things depending on where it appears. The current vector space has no way to represent that difference. Structurally-opposed passages collapse to nearly identical vectors.
- This is not a scale problem. Larger models encode the same one-dimensional position signal with more parameters. A more confident model retrieves the structurally wrong chunk more reliably.
- The consequence propagates forward: the language model receives structurally incorrect context, reasons from it, and produces a confident wrong answer. The failure is invisible in standard benchmarks — BEIR and MTEB contain no structurally-adversarial test pairs.
- The research question: Can a vector space be designed so that structural position — hierarchy depth, section identity, relational role — is encoded as geometry, not discarded? And if so, can the knowledge already compressed inside a pretrained embedding model bootstrap that structural geometry without retraining on raw data?
- HiSE investigates this through five coordinated mechanisms: multi-dimensional structural addressing in pre-attention embedding space, structure-guided attention during fine-tuning, POMDP-grounded theoretical formalization of structural observability, contrastive training with structurally-adversarial hard negatives. HiSE demonstrates that structural geometry is encodable in vector space at the embedding level. The context window problem is solvable from this direction.

Introduction

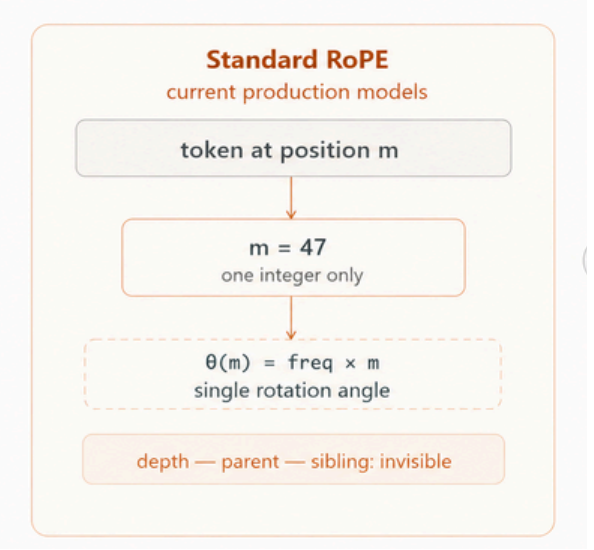
- The global RAG market is projected to grow from $2.33B in 2025 to $81.51B by 2035 (CAGR 42.7%), driven primarily by document retrieval. Over 80% of enterprise data being retrieved is hierarchically structured.
- 71% of organizations use GenAI regularly yet only 17% report meaningful business impact. The bottleneck is not model capability. It is retrieval quality.

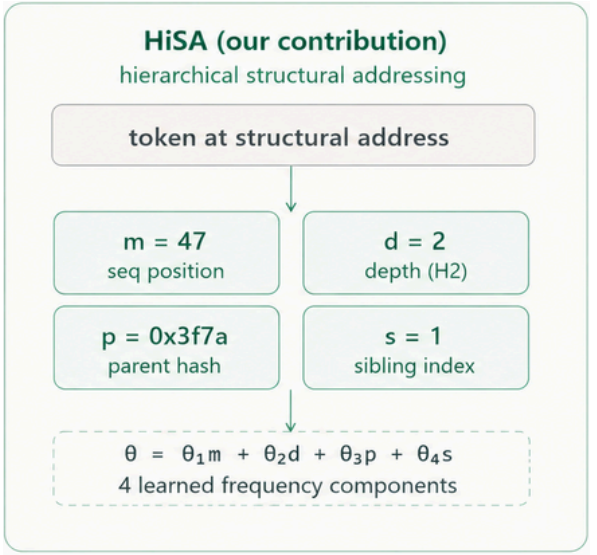
Document → Chunk → Embed (RoPE) → Vector DB → Query Match → LLM

- When the embedding is wrong, the LLM reasons confidently from wrong context.
- {BGE-M3, Qwen3-Embedding, E5-Mistral, Nomic Embed} all encode token position as a single integer via RoPE. Depth, section identity, relational role — none of it enters the vector.
- Naive RAG achieves only 10–40% success rates in enterprise environments. Retrieval failure, not generation failure, is the primary cause of wrong answers in production AI systems.
- Every production RAG pipeline has the same architectural blind spot. A vector space that encodes structural position as geometry replaces context-as-memory with navigation, the model finds what it needs by position, not by re-reading everything.

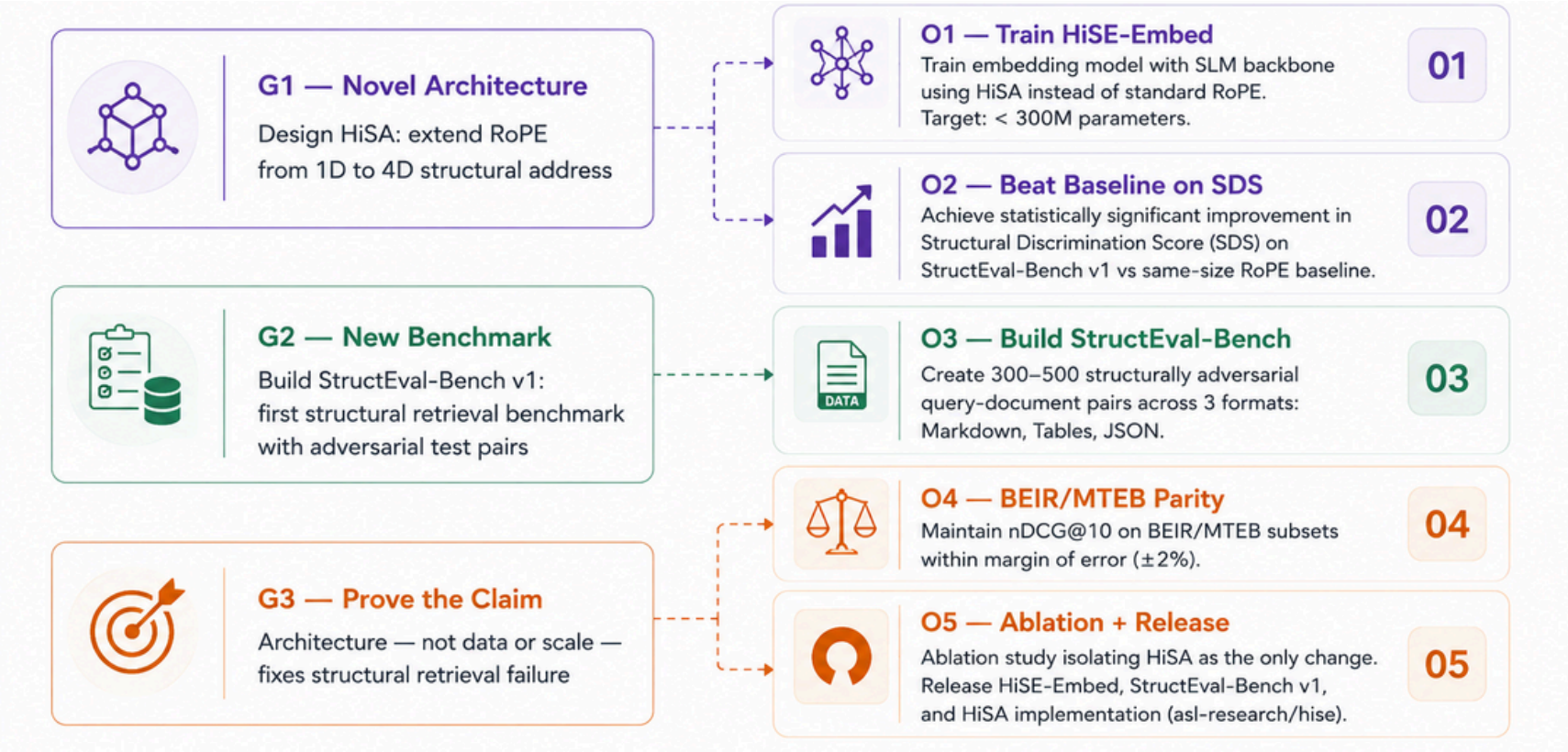
Introduction

- RoPE computes Attention(A,B) ~ f(|pos_A - pos_B|) — sequence distance only. Two tokens under Risks vs Performance are structurally opposed but positionally identical. Depth, parent path, sibling index are mathematically invisible to every production model.
- TUTA, DOM-LM, TabBiN, Struc-EMB — all prompt-level or classification-task workarounds. None touched the positional encoding layer. BEIR and MTEB have zero structurally-adversarial test pairs — the benchmark never exposed the failure, so no model ever optimized against it.
- Enterprise RAG runs over 10K filings, legal contracts, Kubernetes configs, clinical protocols — all hierarchically structured. The higher the structural complexity, the more confidently wrong the retrieval becomes.
- No production model encodes where a token lives in a document. HiSE fixes this at the position encoding layer — encoding structural position as geometry in the vector space itself, and transferring the structural world already latent in pretrained weights into a persistent model of how concepts relate and where they belong.





Goals and Objectives



- Hypothesis: Explicit structural positional encoding enables transformers to distinguish semantically similar but structurally different content.
- Validation Strategy: Verify the hypothesis through controlled benchmarking, ablation studies, and comparison with standard RoPE-based embedding models.

Motivation

Structure Carries Meaning:
Headings, section depth, parent path, and sibling index encode semantics that surface text alone cannot. The same sentence means different things in different structural positions.

Current Models Are Structure-Blind:
Modern embedding architectures preserve token order — not document hierarchy. The architectural blind spot is at the position encoding layer, not in the data or the parameters.

Retrieval Fails Before Generation:
Incorrect embeddings propagate structural errors throughout the RAG pipeline. The language model never gets the chance to reason correctly if the retrieved context is structurally wrong.

HiSE Addresses the Root Cause:
Encode structural position as geometry in the vector space itself, not by scaling parameters, but by correcting the inductive bias where the information is lost.

- Research Gap in Positional Encoding: Existing positional encoding methods model sequential relationships only. Hierarchical structural position — depth, parent path, sibling index — remains unexplored at the encoding layer for dense retrieval. No production model has been built to fill this gap.
- Embedding Quality Determines Retrieval: Dense retrieval systems are fundamentally limited by the quality of their embedding representations. Improving positional encoding directly improves retrieval accuracy.
- The World Model Problem: Language models compensate for absent world structure by consuming larger context windows. This is brute force. If structural position is encoded as geometry in the vector space, the model navigates by position rather than re-reading — breaking the dependency on context scaling.

Literature Review

Paper	Year	What it did	Why it's not enough
TUTA (Microsoft)	2021	Bi-dimensional tree coords for table structure	Classification only - no retrieval training
DOM-LM	2022	Depth + parent + sibling PE in BERT for HTML	HTML-only, additive not rotational, no RAG
Tree-Enhanced CodeBERTa	2025	AST structural PE in CodeBERTa for code	Code/AST only, not a retrieval model
TabBiN (EDBT)	2025	2D tabular coords + visibility matrix	Table-specific, no cross-format, no dense retrieval
SEAL (EMNLP)	2025	Structure-aware contrastive learning	Ain`t introduce a unified rotational positional encode
Struc-EMB (Georgia Tech)	2025	Structural context in LLM-based embedders	Prompt-level workaround, not architectural PE
Rethink RoPE (IMP)	2025	N-dimensional RoPE extension blueprint	Vision/spatial only — not applied to documents
RoPE (IMP)		Encoding relative positional information through rotations	does not model hierarchical structure directly
Attention is all you need	2017	sinusoidal positional encoding for sequence	no dense retrieval objective & linear sequence pos...
Adaptive Quantitative: IDRLA		Introduce Partially observable Markov Decision Process	#

Existing System & its Limitation

Every approach tried to work around the problem. None touched where the problem lives.

System	What it does	Core Limitation
BGE-M3 (570M)	State-of-art dense retrieval, multi-lingual	RoPE-based, 1D position only. nDCG@10: 0.467. No structural signal
Nomic Embed v2 (137M)	Open-source dense retrieval	Same RoPE architecture. Cosine sim ≈ 0.97 on structurally-opposed chunks
Qwen3-Embedding-0.6B	retrieval model for QWEN source model	Linear sequence distance only. Structural info: architecturally invisible
E5-Mistral-7B	Microsoft large-scale embedder	Scaled parameters over broken inductive bias, more confident wrong answer
Struc-EMB (Georgia Tech, 2025)	Structural context via prompt concatenation	Prompt-level workaround — positional encoding layer untouched

Every existing system encodes where a token appears in a sequence. None encodes where a token lives in Semantics.

Scaling data and parameters over a broken inductive bias produces a more confident wrong answer, not a correct one.

Research Gap

No system has ever encoded where data lives in its own structure — only where it appears in a sequence.

1. Every production model encodes position as a single integer via RoPE. Tabular coordinates, JSON key-paths, HTML node depth, Markdown heading hierarchy — all collapsed to m = 47. One number. Every structural dimension discarded.
2. No existing work produces a composite structural address across data formats. TUTA solved tables. DOM-LM solved HTML. Struc-EMB used prompt concatenation. None combined depth, parent identity, and sibling index into a unified rotational signal embedded into the attention mechanism itself — format-agnostic, architecture-level.
3. RoPE operates on a one-dimensional rotational manifold. No system extends this to a multi-dimensional rotational geometry where depth, parent path, and sibling index each occupy independent subspaces with learned frequencies — making structural proximity a geometric property of the vector space, not a statistical outcome of training data.
4. No system applies POMDP to structural address inference. Full document parse at token level is computationally intractable at scale. A Partially Observable Markov Decision Process over structural state gives high-probability address estimation — processing just enough structure to achieve true-negative precision without exhaustive tree computation.
5. No system transfers the structural world already latent in pretrained embedding weights into an explicit geometric representation. The knowledge exists — compressed inside every production model. No work has extracted it and used it to supervise structural fine-tuning without retraining on raw data.

Three fields — structural understanding, data encoding, dense retrieval — each solved a piece. None converged. The composite structural address across all data formats, embedded into the rotational geometry of attention itself, with probabilistic structural inference and world-model knowledge transfer, has never been built.

Proposed System

HiSA- Hierarchical Structural Addressing · From Single Integer to Composite Structural Address

- *Format-Agnostic Structural Addressing: Tabular data, JSON, HTML/XML, Markdown, and complex documents share a limitation: conventional positional encoding primarily represents sequence position \(m\). HiSA introduces a unified structural address \((m,d,p,s)\), representing sequence position, tree depth, parent identity, and sibling index, extracted through format-specific parsers.*
- *Multi-Dimensional Rotational Geometry: Each address dimension occupies an independent subspace with learned rotation frequencies. This aims to preserve structural proximity across distant tokens while distinguishing structure*

$$\mathbb{R}^{768} = \mathbb{R}^{384}_{m} \oplus \mathbb{R}^{154}_{d} \oplus \mathbb{R}^{154}_{p} \oplus \mathbb{R}^{76}_{s}$$

- *Path-Based Composition: HiSA composes rotations along ancestor paths rather than relying on hashes. The goal is to reduce structural collisions and enable algebraic recovery of hierarchical relationships. A rank-64 projection will be investigated to reduce computational cost.*
- *POMDP-Inspired Structural Inference: A partially observable decision framework estimates structural addresses from incomplete document information, aiming to reduce parsing overhead while maintaining retrieval accuracy.*
- *World-Model Knowledge Transfer: A proposed pipeline probes frozen pretrained representations to extract candidate relationships into a structural knowledge graph, providing additional supervision without retraining on the original pretraining corpus.*

Same backbone · same data · same compute · only positional encoding changes — improvement isolates HiSA as the experimental variable.

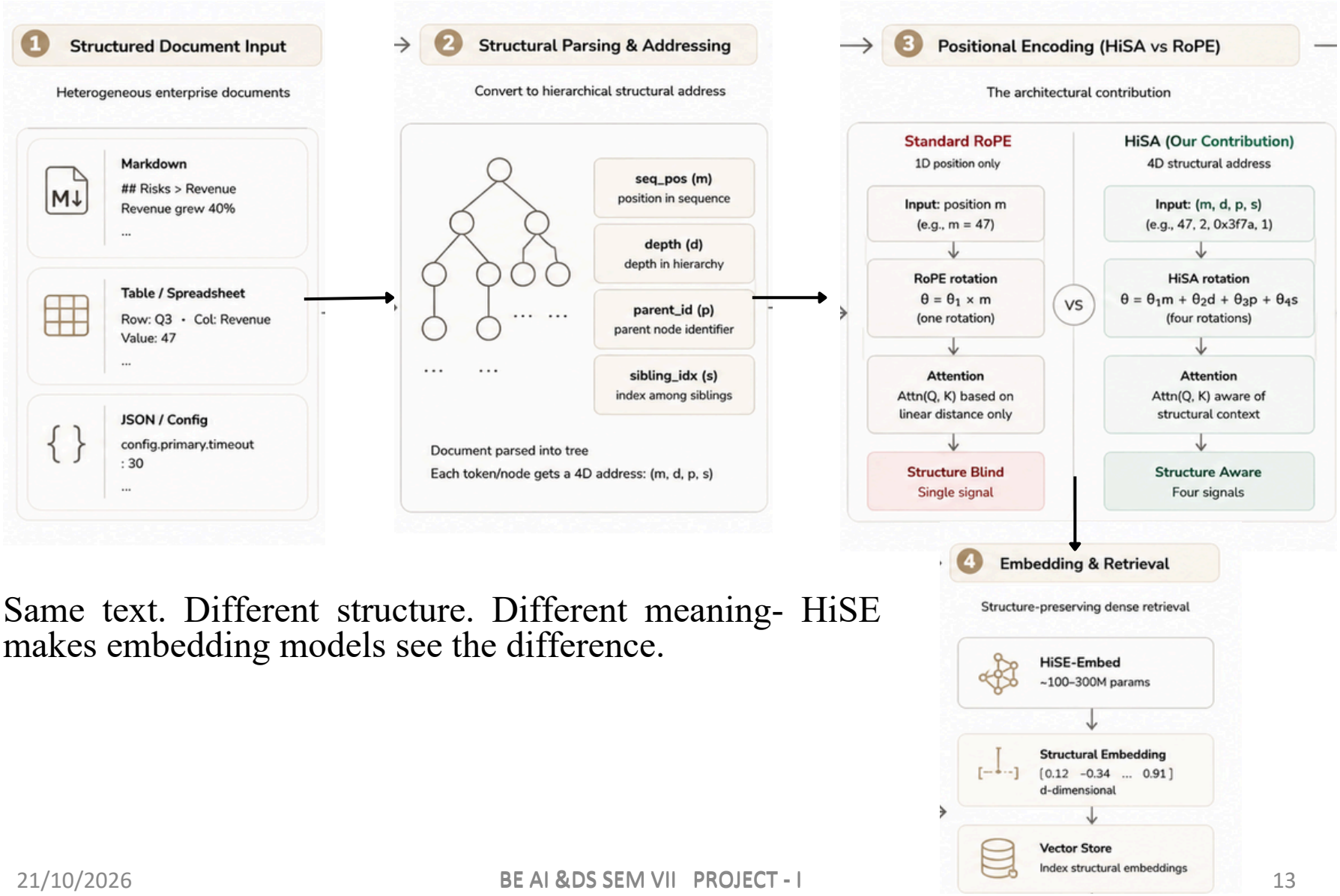
Methodology

From structural failure discovery to hierarchical representation learning, knowledge transfer, and retrieval validation.

1. Failure-Mode Survey: Evaluate BGE-M3, Qwen3-Embedding, and Nomic Embed on structurally adversarial pairs across tables, JSON, HTML/XML, and Markdown. Measure cosine similarity and Retrieval Inversion Rate (RIR) to establish the baseline structural discrimination gap.
2. HiSA Implementation: Implement multi-dimensional structural addressing in a compatible encoder using ((m,d,p,s)). Develop format-specific parsers and investigate independent rotational subspaces, path-based composition, and rank-64 projection. Verify structural distinguishability and equivalence to standard RoPE when structural dimensions are inactive.
3. World-Model Knowledge Transfer: Probe frozen pretrained representations for candidate entity relationships and construct a typed knowledge graph (G). Investigate graph-based structural supervision and BART-encoder query-side structural intent prediction, validating extracted relationships before use.
4. StructEval-Bench v1: Develop 300–500 structurally adversarial query-document pairs across Markdown, tables, and JSON. Define Structural Discrimination Score (SDS) and RIR, and release the benchmark independently on Hugging Face Datasets.
5. Training and Ablation: Fine-tune HiSE-Embed using contrastive learning, structural hard negatives, and curriculum training. Evaluate SGAM attention biases and POMDP-inspired structural inference through controlled ablations to measure each component's contribution.
6. Evaluation and Release: Evaluate structural discrimination and retrieval performance on StructEval-Bench v1, BEIR, and MTEB. Measure SDS, RIR, nDCG@10, computational cost, and statistical significance before releasing HiSE-Embed, StructEval-Bench v1, and the HiSA implementation under asl-research/hise.

The objective is to establish whether hierarchical encoding, structural attention, and pretrained knowledge transfer jointly improve structural retrieval while preserving general semantic performance.

System Architecture



Same text. Different structure. Different meaning- HiSE makes embedding models see the difference.

Modules & Functionalities

Four modules. One pipeline. Structure preserved end to end.

1. **Structural Parser & Addressing Engine:** Format-specific parsers extract structural coordinates from all input data types. Markdown AST walker extracts heading depth and path. Pandas extractor pulls row-column coordinates from tables. JSON recursive traverser extracts key-path depth and parent hash. HTML/XML DOM walker extracts node depth and sibling index. All parsers output a unified (m, d, p, s) tuple — same interface regardless of data format.
2. **HiSA Positional Encoding Layer:** Replaces standard RoPE in the transformer backbone. Accepts (m, d, p, s) tuple per token. Applies four independent learned rotation frequencies producing composite structural address $\theta = \theta_1 m + \theta_2 d + \theta_3 p + \theta_4 s$. Embedded directly into $Q \cdot K$ attention computation. 96 new scalar parameters total.
3. **POMDP Structural State Estimator:** Estimates composite structural address probabilistically when full document parse is computationally intractable. Maintains a belief state over structural position — depth, parent identity, sibling index — updating with each observed token. Processes just enough structure to achieve true-negative level precision without exhaustive tree computation at scale.
4. **Contrastive Training & Evaluation Pipeline:** Trains HiSE-Embed via InfoNCE loss with two negative categories: standard semantic hard negatives and structurally-adversarial hard negatives — identical surface text, structurally-opposed ground truth. Evaluates on StructEval-Bench v1 (SDS, RIR metrics) and BEIR/MTEB (nDCG@10) for general retrieval parity confirmation.

Each module is independently testable. Each has a defined input, output, and success criterion.

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Technology & Requirements

- **Software Requirements:**

1. PyTorch 2.x · CUDA 12.8 · HuggingFace Transformers
2. Sentence-Transformers · custom HiSA module replacing RoPE
3. FAISS vector indexing · BEIR · MTEB · StructEval-Bench v1
4. Python-Markdown · pandas · JSON key-path parser
5. Weights & Biases · HuggingFace Hub for open-source release

- **Hardware Requirements:**

1. Cuda Core Processing systems · Unifiled Video Memory · primary training · ASL server
2. Local System Resources · Primary Memory · local development
3. High BandWidth system processor
4. 500GB SSD · checkpoints and embedding index storage
5. Random Access Memory · corpus loading and FAISS operations

Software:
All development and training containerized via Docker on ASL research infrastructure · reproducible builds guaranteed across environments.
Hardware:
Primary training on NVIDIA RTX (Blackwell architecture) · prototype validation on RTX · both locally and hosted dependency.

Expected outcome

Three research artifacts. One structure-aware retrieval framework.

1. HiSE-Embed (The Model): A dense retrieval model under 300M parameters integrating HiSA, SGAM, and contrastive learning with structurally adversarial hard negatives. Supports Markdown, tables, JSON, HTML/XML, and complex documents. Target: open-source release on Hugging Face Hub under asl-research/hise-embed (Apache 2.0).
2. StructEval-Bench v1 (The Benchmark): A benchmark of 300–500 structurally adversarial query-document pairs spanning hierarchical Markdown, tabular relationships, and nested JSON. Introduces Structural Discrimination Score (SDS) and Retrieval Inversion Rate (RIR). Target: release on Hugging Face Datasets (CC BY 4.0).
3. Validated Structural Retrieval Framework (The Evidence): Controlled experiments and ablations evaluating structural encoding, SGAM, knowledge transfer, and structural inference against established embedding baselines.

Metric	Target	Meaning
SDS on StructEval-Bench v1	Statistically significant improvement ((p<0.05))	Better structural discrimination urally discriminates
RIR on adversarial pairs	Reduction against baseline	Fewer structurally incorrect top-ranked results
nDCG@10 on BEIR/MTEB	Within ±2% of baseline	Preserve general retrieval quality
Parameter count	≤ 300M parameters	Practical deployment and fine-tuning

Project Timeline

Phase	Timeline	Work	Success Criterion
Failure Survey	**Month 1–2**	Quantify structural discrimination gap on BGE-M3, Qwen3, Nomic across tabular, JSON, HTML, Markdown adversarial pairs	**Cosine sim ≥ 0.90 on structurally-opposed pairs. RIR baseline established**
HiSA Build	**Month 2-3**	Replace RoPE with 4D structural address. Build all format parsers outputting unified (m, d, p, s). Implement POMDP estimator	**Degenerate case: d=0, p=0, s=0 → identical output to standard RoPE**
Benchmark	**Month 3-4**	Build StructEval-Bench v1 — 300–500 adversarial pairs. Define SDS and RIR metrics. Release on HuggingFace Datasets	**Benchmark live with annotation guidelines**
Training	**Month 4-5**	Train HiSE-Embed with structurally-adversarial hard negatives. Run controlled ablation — only positional encoding changes	**SDS improvement statistically significant ($p < 0.05$) vs RoPE baseline**
Eval + Release	**Month 6**	Evaluate on BEIR/MTEB. Release HiSE-Embed, StructEval-Bench v1, HiSA code. Submit to ICNLP 2027	**nDCG@10 within ±2%. Submission by Nov 30, 2026**

Conclusion & Future Scope

Conclusion :

- HiSA: A proposed multi-dimensional positional encoding framework integrating sequence position, depth, parent identity, and sibling index across structured and complex documents.
- HiSE-Embed: A contrastively trained dense retrieval model using structural hard negatives and SGAM to improve structure-sensitive semantic retrieval.
- Knowledge Transfer: Investigates pretrained representation probing, validated knowledge graphs, and query-side structural intent prediction.
- StructEval-Bench v1: A structural retrieval benchmark introducing SDS and RIR to quantify structural discrimination and retrieval failures.
- Empirical Validation: Controlled ablations assess encoding, attention, and knowledge-transfer contributions while measuring general retrieval performance.

Future Scope:

- Generative LLMs: Extend hierarchical positional encoding to structure-aware generation and long-context reasoning.
- Cross-Lingual Retrieval: Evaluate hierarchical structure encoding across Indic languages and multilingual documents.
- Advanced Knowledge Transfer: Explore graph-guided structural inference and efficient address prediction for large, complex documents.
- Benchmark Expansion: Extend StructEval-Bench to legal, clinical, and financial documents with expert-annotated structural relationships.

HiSE investigates a fundamental limitation of sequence-based representations: linear proximity does not necessarily capture structural relationships within data.

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Thank You !